



Laboratory assessment and durability performance of vinyl-ester, polyester, and epoxy glass-FRP bars for concrete structures



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ABSTRACT

In the last decade, noncorrosive glass fiber-reinforced-polymer (GFRP) bars have become more widely accepted as cost-effective alternatives to steel bars in many applications for concrete structures (bridges, parking garages, and water tanks). Also, these reinforcing bars are valuable for temporary concrete structures such as soft-eyes in tunneling works. The cost of the GFRP bars can be optimized considering the type of resin according the application. Yet limited research seems to have investigated the durability of GFRP bars manufactured with different types of resin. In this study, the physical and mechanical properties of GFRP bars made with vinyl-ester, isophthalic polyester, or epoxy resins were evaluated first. The long-term performance of these bars under alkaline exposure simulating a concrete environment was then assessed in accordance with ASTM D7705. The alkaline exposure consisted in immersing the bars in an alkaline solution for 1000, 3000 and 5000 h at elevated temperature (60 °C) to accelerate the effects. Subsequently, the bar properties were assessed and compared with the values obtained on unconditioned reference specimens. The test results reveal that the vinyl-ester and epoxy GFRP bars had the best physical and mechanical properties and lowest degradation rate after conditioning in alkaline solution, while the polyester GFRP bars evidenced the lowest physical and mechanical properties and exhibited significant degradation of physical and mechanical properties after conditioning.

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1. Introduction

Fiber-reinforced-polymer (FRP) bars have been well accepted as internal and external reinforcement for concrete structures (ACI 440.1R [4]; [6,28,48]). This reinforcing material offers better resistance to environmental agents as well as high stiffness-to-weight and strength-to-weight ratios when compared with conventional construction materials such as steel. Extensive research and development efforts have demonstrated that FRP bars are effective reinforcement in concrete members subject to bending [44], shear [5] and [7], compression [45], and impact [38]. Material

specifications and design guidelines (ACI 440.6M [3]; CAN/CSA S807 [32]) have also been developed to encourage the construction industry to use FRP bars. This has resulted in many demonstration projects and field applications, such as bridges [27], parking garages [25], water-treatment plants [47], bridge barriers [36], concrete pavement [26], and jetties [42].

Different types of fibers are used in manufacturing FRP bars such as carbon, glass, aramid, and basalt. Many studies have been carried out on the performance and use of FRP bars made with these different fibers, providing good insight into their physical and mechanical properties as well as their durability characteristics [1,8,10,11,22,28,31,39–41,57]. Glass is the most commonly used fiber type in manufacturing FRP bars due to their relatively low comparative cost [4]. Similarly, [34] highlighted the importance of the resin system used in manufacturing FRP bars to achieve the desired mechanical properties and durability characteristics. The resin system is important as it acts as a matrix bonding the fibers

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together and spreading the load applied to the composite between each of the individual fibers. The resin system also protects the fibers from abrasion and impact damage as well as from severe environmental conditions—such as water, salts, and alkalis—which affect the durability of FRP products [56]. A deterioration of this interface reduces the transfer of the loads between fibers and thus weakens the composite materials [9]. The interface between the fiber and matrix is a nonhomogeneous region about 1 μm thick. This layer is weakly bonded and most vulnerable to deterioration. The three dominant deterioration mechanisms include matrix osmotic cracking, interfacial debonding, and delamination [35]. Moisture diffusion into FRP composites could be influenced by the material's anisotropic and heterogeneous character. Along with diffusion into the matrix, wicking through the fiber/matrix interface in the fiber direction could be a predominant mechanism of moisture ingress [13]. Nonvisible dissociation between fibers and matrix could lead to rapid losses of interfacial shear strength [14,37]. Unfortunately, limited research attention has been paid to the effect of the resin-system type on the physical and mechanical properties as well as the durability characteristics of GFRP bars.

Most of the glass-fiber-reinforced polymer (GFRP) bars available are manufactured with E or ECR glass fibers that are normally wetted with a thermosetting resin such as epoxy or vinyl ester. Numerous studies have investigated FRP bars made with vinyl-ester resin to determine the effect of environmental conditions (water, salts, alkalis) on their physical and mechanical properties [24,29–31,50,52,53,58,59]. Similarly, [8,12,55]; and [28] are some of the numerous researchers who have investigated the durability performance of FRP bars made with epoxy resins. GFRP bars made with these resin systems are the most commonly used as reinforcement for concrete structures given their high performance and very good durability characteristics. Studies into the behavior of fiber-reinforced isophthalic polyester-resin composites have primarily addressed industrial and nonstructural products such as natural-fiber composites [43]. GFRP bars manufactured with isophthalic polyester resin are normally used for temporary structures such as soft-eyes in underground excavations and tunneling works [54]. In these proprietary applications, GFRP-bar durability is not a concern. The key advantage of GFRP bars is the low cost of polyester resin and the fact that GFRP bars can be cut without damaging the drilling equipment's cutter heads. Comparisons performed by some researchers [2,14,15] have indicated that the matrix formed by vinyl ester, which contains many fewer ester units compared to polyester, experiences very little deterioration caused by hydroxyl ions compared to a polyester matrix. As a result, CSA S807 [32] classifies isophthalic polyester-based GFRP bars as having moderate durability (D2), while classifying epoxy- and vinyl-ester-based GFRP bars as having high durability (D1). Obviously, these classifications were established based on the results obtained by different researchers on GFRP bars manufactured with a specific resin system, i.e., either vinyl esters or epoxies (with very few studies on isophthalic polyesters). Consequently, no sound generalizations can be made. Clearly, a single approach is needed to confirm that GFRP bars manufactured with different types of resin will have different physical and mechanical properties as well as different durability characteristics, therefore providing for direct comparison of these important properties.

This paper presents an experimental investigation aimed at assessing and comparing the physical and mechanical properties of three different types of GFRP bars made with vinyl-ester, isophthalic polyester, or epoxy resins. The tests findings on the long-term durability of these bars conditioned in an alkaline solution simulating a moist concrete environment at high temperature are also presented. The aim is to further understanding of the various resin options available for GFRP bars and their associated

behavioral characteristics, yielding useful information about materials specifications and design standards.

2. Experimental-program outline

This experimental investigation was conducted on three different GFRP bars that were 12 mm in diameter: glass/polyester, glass/vinyl-ester, and glass/epoxy as shown in Fig. 1. The experimental work was divided into three phases. Phase I included the determination of physical properties, which were compared to those obtained after conditioning. Phase II included mechanical characterization, including transverse shear strength, flexural strength, flexural modulus of elasticity (stiffness), and apparent horizontal shear strength (interlaminar-shear strength). The test results also served as references for residual strength after conditioning. Phase III included a preliminary durability assessment and long-term performance assessment of the GFRP bars immersed in alkaline solutions simulating concrete pore solution at 60 °C for different times up to 5000 h. The changes in the physical and mechanical characteristics were assessed by comparing the characteristics of the conditioned bars to those of the reference bars from Phases I and II. The effects of conditioning on the glass transition temperature (T_g) and chemical composition of the materials were also assessed with differential scanning calorimetry (DSC) and Fourier transform infrared spectroscopy (FTIR), respectively. In addition, bar microstructure was investigated using scanning electron microscopy (SEM) for both conditioned and unconditioned bars to assess changes and/or degradation.

3. Material properties and test specimens

The glass/polyester, glass/vinyl-ester, and glass/epoxy FRP bars were manufactured with continuous glass fibers impregnated in polyester, vinyl-ester, or epoxy resins using the pultrusion process. Table 1 lists the typical properties of these thermosetting-resin systems as reported by Ref. [23]. The three types of GFRP bars (Fig. 1) were manufactured by Firep International AG (Switzerland) using the same fabrication process and equipment, same glass fiber, and same additives to ensure bar consistency. The GFRP bars had a nominal diameter of 12 mm and were deformed with helical wrapping (Fig. 1). The nominal cross-sectional area of the three GFRP bars was 113 mm², as reported by the manufacturer. The mechanical properties reported herein were calculated using the nominal cross-sectional area.

For this study, the GFRP bars were provided in 170 and 240 mm lengths so that the transverse shear-strength test and flexural test could be performed according to ASTM D7617 [19], and ASTM D4476 [17], respectively. In addition, some specimens were cut into



Fig. 1. Tested GFRP bars.

Table 1
Typical properties of the thermosetting resins [23].

Property	Resin system		
	Polyester	Vinyl-ester	Epoxy
Glass transition temperature (T_g), °C	100	110	120
Tensile modulus, GPa	4.0	3.5	3.0
Tensile strength, MPa	65	82	90
Elongation at break, %	2.5	6.0	8.0

83 mm lengths so that the short-beam shear test could be performed according to ASTM D4475 [16] on the three types of GFRP bars.

4. Testing, results, and discussion

4.1. Phase I: physical characterization

Physical properties for the reference (unconditioned) GFRP bars were determined according to [3] and [32] requirements, including: (1) fiber content, (2) moisture absorption, (3) cure ratio, and (4) glass transition temperature.

4.1.1. Fiber content

Glass fiber content was determined by thermogravimetry according to ASTM E1131. A very small piece of material (a few tenths milligrams) was cut from the center of the bar, placed in platinum crucible and then heated up to 550 °C under inert atmosphere. The weight loss (W_L) has been recorded at a temperature equal to 550 °C. Since the material only contains carbon fibers and resin, fiber content by weight was then calculated according to the following equation:

$$\text{Fiber content by weight} = 100 \cdot (W_T - W_L) / W_T \quad (1)$$

where, W_T is the total weight before burn off.

4.1.2. Water-immersion test

The moisture uptake at saturation of the GFRP bars was determined according to ASTM D570, except that the immersion was in tap water instead of distilled water. Three 50 mm long specimens were cut, dried, and weighed prior to immersion in water at 50 °C for three weeks. The samples were removed from the water after three weeks, surface dried, and weighed.

The water content at saturation in weight percent (W_s) was calculated using Equation 2.

$$W_s = 100 \cdot (P_s - P_d) / P_d \quad (2)$$

where P_s and P_d are the sample weights in the saturated and dry states, respectively.

4.1.3. Cure ratio

Cure ratio was determined according to ASTM D5028 and Ref. [32]. The enthalpy of polymerization of the sample was measured by DSC and compared to the enthalpy of polymerization of pure resin, taking into account the weight percentage of resin in the matrix. Thirty to fifty milligrams of sample were accurately weighed and placed in an aluminum crucible. The samples were then heated from room temperature to 200 °C at a heating rate of 20 °C/min, and the area of the peak of polymerization was calculated. The measurement was carried out on 3 specimens.

4.1.4. Transverse coefficient of thermal expansion

The transverse coefficient of thermal expansion was calculated according to ASTM E1131-08 (2014). Nine specimens of each type of GFRP bars were tested. The measurements were conducted between −30 °C and 60 °C at a heating rate of 3 °C. A TA Q400 thermomechanical analyzer was used. Cryogenic equipment (liquid nitrogen) was used to reach subzero temperatures. The results show that the coefficients of thermal expansion for the different bar diameters fell between $20.5 \times 10^{-6}/^{\circ}\text{C}$ and $22.0 \times 10^{-6}/^{\circ}\text{C}$, which is only half of the limit of $40.0 \times 10^{-6}/^{\circ}\text{C}$ specified in CSA 807 [32].

Table 2 lists the physical properties of the unconditioned GFRP bars, where the glass transition temperature (T_g) was determined with differential scanning calorimetry (DSC) [ASTM D3418 [21]] (see Fig. 2).

As shown in Table 1, the glass/polyester and glass/epoxy FRP bars had approximately the same fiber content (78.8% and 79.4% by weight, respectively), while the glass/vinyl-ester FRP had the highest fiber-content ratio (83.9% by weight). The average cure ratios and transverse coefficients of thermal expansion of the tested bars were around 99.0 ± 1.0 and $19.25 \pm 1.55 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$, respectively, without significant differences between the three types of bars tested. On the other hand, significant differences were observed for T_g and moisture uptake. The vinyl-ester and polyester GFRP bars returned T_g values of 113 °C and 93.0 °C, respectively, while the epoxy GFRP bars had a T_g value of 126 °C. Similarly, the vinyl-ester and polyester GFRP bars had water uptake ratios of 0.63% and 1.15%, respectively, while the epoxy GFRP bars had a moisture-uptake ratio of 0.23%. The limits of water absorption of the bars at saturation were <1% and <0.75% for high and medium durability, respectively, as recommended in CSA S807 [32]. The measured water absorption of the polyester GFRP bars was slightly higher than these limits, probably due to the resin-rich deformation pattern on the bar surface, which absorbed most of the moisture.

4.2. Phase II: mechanical characterization

The mechanical characterization included testing of representative GFRP bars to determine their transverse shear strength in accordance with ASTM D7617 [19]; interlaminar shear (short-beam test) in accordance with ASTM D4475 [16]; and flexural strength and flexural modulus of elasticity in accordance with ASTM D4476 [17]. These tests were selected as they are primarily related to resin properties and can provide a comparative performance assessment of the three GFRP bars tested herein. Figs. 3–5 show the mechanical characterization tests and Table 3 lists the results. The following sections provide brief descriptions and interpretation of the results.

4.2.1. Transverse-shear strength test

Transverse shear is the major structural force on dowels in jointed pavements or on stirrups in concrete beams. Transverse-shear tests were conducted according to ASTM D7617 (ASTM

Table 2
Physical properties of the reference GFRP bars.

Property	GFRP bar type		
	Polyester	Vinyl-ester	Epoxy
Fiber content by weight (%)	78.8	83.9	79.4
Cure ratio (%)	98.1	99.1	100
Transverse CTE, ($\times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$)	20.8	17.7	19.7
Glass transition temperature, T_g (°C)	93.0	113	126
Moisture uptake (%)	1.15	0.63	0.23

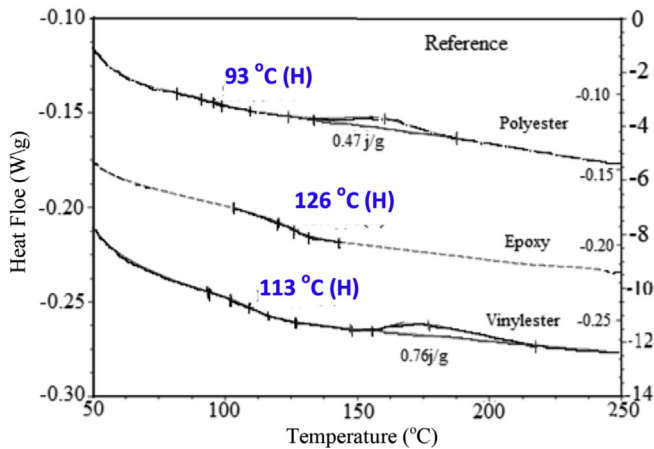


Fig. 2. DSC scans for glass transition temperature (T_g).

2011) to characterize the tested bars. The setup consisted of a $230 \times 100 \times 110$ mm steel base equipped with lower blades spaced at 50 mm face to face, allowing for the double transverse-shear failure of the specimen caused by an upper blade, as shown in Fig. 3. For each type of bar tested, six unconditioned specimens measuring 170 mm in length were tested under laboratory conditions on an MTS 810 (MTS Systems Corporation, Eden Prairie, Minneapolis) testing machine equipped with a 500 kN load cell. A displacement-controlled rate of 1.3 mm/min was used during the test, which yielded between 30 and 60 MPa/min until specimen

failure. The loading was done without subjecting the test specimens to any shock. The transverse-shear strength was calculated with Eq. (1)

$$\tau_u = \frac{P_s}{2A} \quad (1)$$

where τ_u = transverse-shear strength (MPa); P_s = failure load (N); and A = bar cross-sectional area (mm^2).

Table 3 shows that the transverse-shear strengths of the polyester and vinyl-ester GFRP bars were 250 ± 33 and 258 ± 32 MPa, respectively. The epoxy GFRP bars had the highest value of transverse-shear strength (270 ± 45 MPa). It is worth mentioning, however, that, although the resin delivers most of the transverse-shear strength, the fiber and the fiber/resin interface also play a role [49]. The ratios between the shear strengths of the polyester and vinyl-ester GFRP bars and that of epoxy bars were 93% and 96%, respectively. The results indicate that the epoxy resin yielded higher transverse-shear strength than the polyester and vinyl-ester resin, although the standard deviation was high. Moreover, these values meet CSA requirements [32], which specify a minimum transverse-shear strength of 160 MPa for GFRP bars.

4.2.2. Three-point flexural test

Flexural testing is especially useful for quality control and specification purposes. Flexural properties may vary with specimen diameter, temperature, weather conditions, and differences in rates of straining. The flexural properties obtained with this test method—ASTM D4476 [17]—cannot be used for design purposes but are appropriate for the comparative testing of composite materials. The

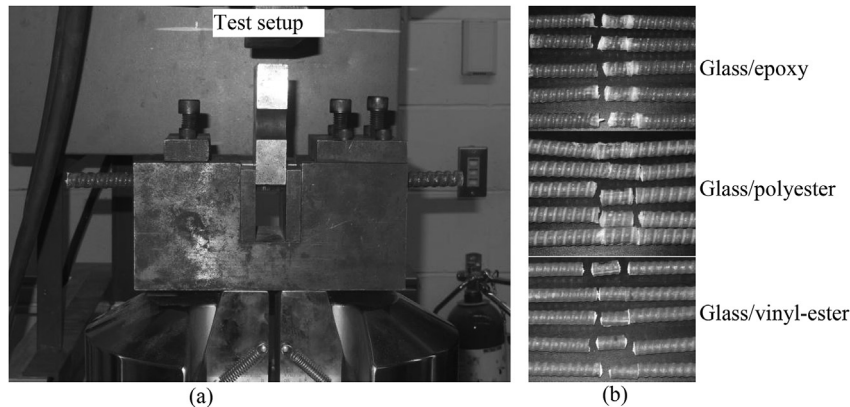


Fig. 3. Setup for transverse-shear test and typical shear failure mode: (a) test setup; (b) failure mode.

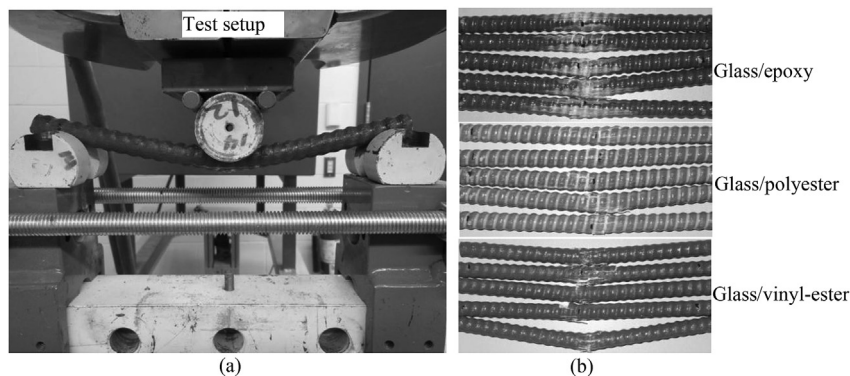


Fig. 4. Setup for flexural testing and typical failure mode: (a) test setup; (b) failure mode.



Fig. 5. Setup for short-beam testing and typical failure mode: (a) test setup; (b) failure mode.

Table 3

Mechanical properties of the reference GFRP bars.

Bar type	τ_u (MPa)	f_u (MPa)	E (GPa)	ϵ_u (%)	S_u (MPa)
Glass/polyester	250 ± 33	1150 ± 59	56.9 ± 2.4	2.02 ± 0.16	47.2 ± 0.4
Glass/vinyl-ester	258 ± 32	1432 ± 75	66.3 ± 2.2	2.16 ± 0.089	64.8 ± 4.5
Glass/epoxy	270 ± 45	1573 ± 135	61.8 ± 1.5	2.54 ± 0.015	77.4 ± 2.7

test was conducted on specimens 240 mm long over a simply supported span equal to 20 times the bar diameter, as shown in Fig. 4. Six unconditioned specimens were tested under laboratory conditions as references for each type on an MTS 810 testing machine equipped with a 500 kN load cell. The specimens were loaded at mid-span with a circular nose; the specimen ends rested on two circular supports that allowed the specimens to bend. A displacement-controlled rate of 3.0 mm/min was used during the test. The rate of loading was done without subjecting the test specimen to any shock. The applied load and deflection were recorded during the test on a data-acquisition system monitored by a computer.

The flexural strength of tested FRP specimens was calculated with Eq. (2). The flexural modulus of elasticity (stiffness) is the ratio, within the elastic limit, of stress to corresponding strain. It was calculated with Eq. (3)

$$f_u = PLC / (4I) \quad (2)$$

$$E = PL^3 / (48IY) \quad (3)$$

where f_u = flexural strength in the outer fibers at mid-span (N/mm²); P = failure load (N); L = clear span (mm); I = moment of inertia (mm⁴); C = distance from the centroid to the extremities (mm); E = flexural modulus of elasticity in bending (N/mm²); and Y = mid-span deflection at load P (mm).

The maximum outer fiber strain (ϵ_u) was calculated from Eq. (4)

$$\epsilon_u = f_u / E \quad (4)$$

Table 3 provides the three-point flexural strength, flexural modulus of elasticity, and ultimate outer-fiber strain of the tested GFRP bars. The elastic behavior of all the specimens was maintained until flexural failure, at which point the specimens failed due to compression in the top fibers, as shown in Fig. 4. The polyester GFRP bars showed the lowest flexural strength (1150 ± 59 MPa), while the epoxy GFRP bars recorded the highest (1573 ± 135 MPa). The vinyl-ester GFRP bars recorded a flexural strength of 1432 ± 75 MPa. The vinyl-ester and epoxy GFRP bars, however,

evidenced no significant differences in flexural modulus of elasticity (66.3 and 61.8 GPa, respectively). Lastly, the flexural modulus of elasticity of the polyester resin was lower than that of the vinyl-ester and epoxy resin (86% and 92% of the vinyl-ester and epoxy GFRP bars, respectively). The lower flexural strength and modulus of the polyester GFRP bars is expected since the polyester had the lowest mechanical properties of the thermosetting resins considered (Table 1). [34] pointed out that the resin system significantly affected the mechanical properties of FRP bars due to the efficiency of the stress transfer among the fibers.

4.2.3. Short-beam shear test

In FRP bars manufactured with a pultrusion process in which the fibers are arranged unidirectionally and bonded with the polymer matrix, the horizontal stresses would be more conducive to inducing interface degradation than transverse-shear stresses [51]. The short-beam shear test was conducted according to ASTM D4475 [16] on six specimens of each type of GFRP bar in order to calculate the interlaminar-shear strength, which is governed by the fiber–matrix interface. The tests were carried out with a 500 kN MTS 810 testing machine. The distance between the shear planes was set to 7 times the nominal diameter of the FRP bars. Fig. 5 shows the test setup and typical modes of failure of the tested specimens. A displacement-controlled rate of 1.3 mm/min was employed during the test. The applied load was recorded with a computer-monitored data-acquisition system.

The interlaminar-shear strength, S_u , of the GFRP bars was calculated from Eq. (5)

$$S_u = 0.849P / d^2 \quad (5)$$

where S_u = interlaminar-shear strength (MPa); P = shear failure load (N); and d = bar diameter (mm).

The short-beam shear test revealed that the epoxy GFRP bars had the highest interlaminar-shear strength (77.4 ± 2.7 MPa), followed by the vinyl-ester GFRP bars (64.8 ± 4.5 MPa) and the polyester GFRP bars (47.2 ± 0.4 MPa). The results confirm that the interface between the glass fibers and polyester resin was not as strong as that within the vinyl-ester and epoxy GFRP bars. Table 3

shows the apparent horizontal shear strength of the tested GFRP bars. It is worth mentioning that the high values of the interlaminar-shear strength reveal a strong interface between the resins and reinforcing fibers, which will be clarified in the SEM analysis to follow.

4.3. Phase III: durability study in alkaline solution

4.3.1. Conditioning of the GFRP bars in alkaline solution

Accelerated aging tests were conducted in accordance with ASTM D7705 [20]. The conditioning of the bars included the combined exposure to a harsh alkaline environment and elevated temperature. Immersion in an aqueous media (alkaline solution) at high temperature accelerates degradation. The alkaline solution was prepared with calcium hydroxide, potassium hydroxide, and sodium hydroxide (118.5 g of $\text{Ca}(\text{OH})_2$ + 0.9 g of NaOH + 4.2 g of KOH in 1 L of deionized water) according to ASTM D7705 and CSA S806 [33]. The pH of the alkali solution was 12.8. The three types of FRP bars—glass/polyester, glass/vinyl-ester, and glass/epoxy—were immersed in this solution at 60 °C for up to 5000 h. The timing of conditioning started once the solution had reached the prescribed temperature. [53] reported that the degradation reaction rate increased almost linearly between room temperature and 50 °C, whereas the increase was exponential at higher temperatures (over 60 °C). Therefore, to avoid any thermal degradation, the maximum conditioning temperature used in this study was 60 °C, as specified in ASTM D7705 [20].

The GFRP specimens were placed in hermetically sealed stainless-steel containers to prevent excessive evaporation and the reaction of atmospheric CO_2 with calcium hydroxide. The containers were placed in an environmental chamber adjusted to the prescribed temperature (60 °C) under isothermal conditions. The bars were weighed and their diameters measured throughout the conditioning period to monitor water absorption and eventually characterize the mass and diameter changes. Observation revealed no changes in diameter during conditioning. Six specimens of each type of FRP bar were removed from the solution and tested to determine their transverse-shear strength, interlaminar-shear strength, flexural properties, and physical properties after 1,000, 3,000, and 5000 h at 60 °C. Durability was assessed using tests for transverse-shear strength [ASTM D7617 [19]], interlaminar shear (short-beam test) [ASTM D4475 [16]], flexural strength, and flexural modulus of elasticity [ASTM D4476 [17]]. Degradation mechanisms in FRP bars are typically denoted as (1) fiber dominated; (2) matrix dominated; and (3) fiber–matrix interface dominated or combined mechanisms. Changes in mechanical properties determined by these tests are indicators of the three specific modes of degradation of the FRP constituent materials given earlier: fibers (flexural tests), resin (transverse and short-beam shear tests), and interface region (short-beam shear and flexural tests). The results for the conditioned specimens were compared to those of the reference ones.

4.3.2. Transverse-shear strength of the conditioned GFRP bars

Table 4 shows the transverse-shear strength and strength-retention ratios of the tested bars after 1,000, 3,000, and 5000 h of immersion in the alkaline solution at 60 °C. Table 4 indicates that the polyester GFRP bars were highly affected by accelerated aging with a transverse-shear strength reduction of 22.5% after 5000 h of immersion, while the vinyl-ester and epoxy bars had transverse-shear strength reductions of 15.9% and 11%, respectively. Fig. 6(a) shows the effect of the alkaline solution on the transverse shear strength after different exposure times. Contrary to the polyester bars, the vinyl-ester and epoxy GFRP bars exhibited no significant reductions in the early stages (less than 3000 h).

4.3.3. Flexural strength of the conditioned FRP bars

Table 4 provides the flexural strength and strength-retention ratios of the tested FRP bars after 1,000, 3,000, and 5000 h of immersion. Both the polyester and epoxy GFRP bars had similar flexural-strength reductions after 5000 h (25 and 23%, respectively), while the vinyl-ester GFRP bars showed a lower reduction of 17%. These observations confirm that the bond between the GFRP fibers and polyester resin—before and after conditioning—was lower than that between the glass fibers in the vinyl-ester or epoxy resin. Consequently, debonding occurring at the fiber–matrix interface caused the fibers to separate from the resin. Fig. 6(b) shows the effect of the alkaline solution on flexural strength. The lowest reduction rate was observed with the vinyl-ester GFRP bars, which yielded the lowest degradation at the interface. The high degradation of the epoxy GFRP bars after 1000 h of conditioning resulted from the ingress of the alkaline solution through the initial voids. The polyester GFRP bars, however, returned an almost steady degradation rate between 1000 and 5000 h.

4.3.4. Flexural modulus of elasticity of the conditioned GFRP bars

Table 4 gives the flexural modulus of elasticity and the retention ratio of the tested FRP bars after 1,000, 3,000, and 5000 h of immersion. The three bar types had no significant differences in flexural modulus of elasticity after 5000 h. The reduction ranged from 10.7% to 12.6% in comparison to the references. Fig. 6(c) illustrates the effect of the alkaline solution on the flexural modulus of elasticity, with all types of bar specimens exhibiting a steady reduction rate.

4.3.5. Interlaminar-shear strength of the conditioned GFRP bars

Table 4 also shows the apparent horizontal shear (interlaminar shear) strength and strength-retention ratios of the tested FRP bars after 1,000, 3,000, and 5000 h of immersion. As for flexural testing, the vinyl-ester and epoxy GFRP bars offered excellent stability and durability after immersion in the alkaline solution, followed by the polyester GFRP bars. The reduction ratios for the vinyl-ester, epoxy, and polyester GFRP bars after 5000 h were 13%, 13%, and 21%, respectively. Again, this observation confirms the strong fiber–resin interface in the vinyl-ester GFRP bars, followed by the epoxy and polyester GFRP bars. As evidenced from these results, the fiber–resin interface stands out as one of the most important issues in manufacturing glass FRP. Fig. 6(d) shows the effect of the alkaline solution on the interlaminar-shear strength, with the vinyl-ester GFRP bars exhibiting the lowest rate of degradation. Interestingly, the 21% reduction in the interlaminar-shear strength of the polyester GFRP bars in this study is significantly lower than with the polyester E-glass composite rods tested by Ref. [46]; who observed a more than 90% reduction in interlaminar-shear strength. This indicates that the development of new material systems and advanced manufacturing methods now yield high-quality FRP bar products.

4.3.6. Microstructural analysis of the reference and conditioned GFRP bars

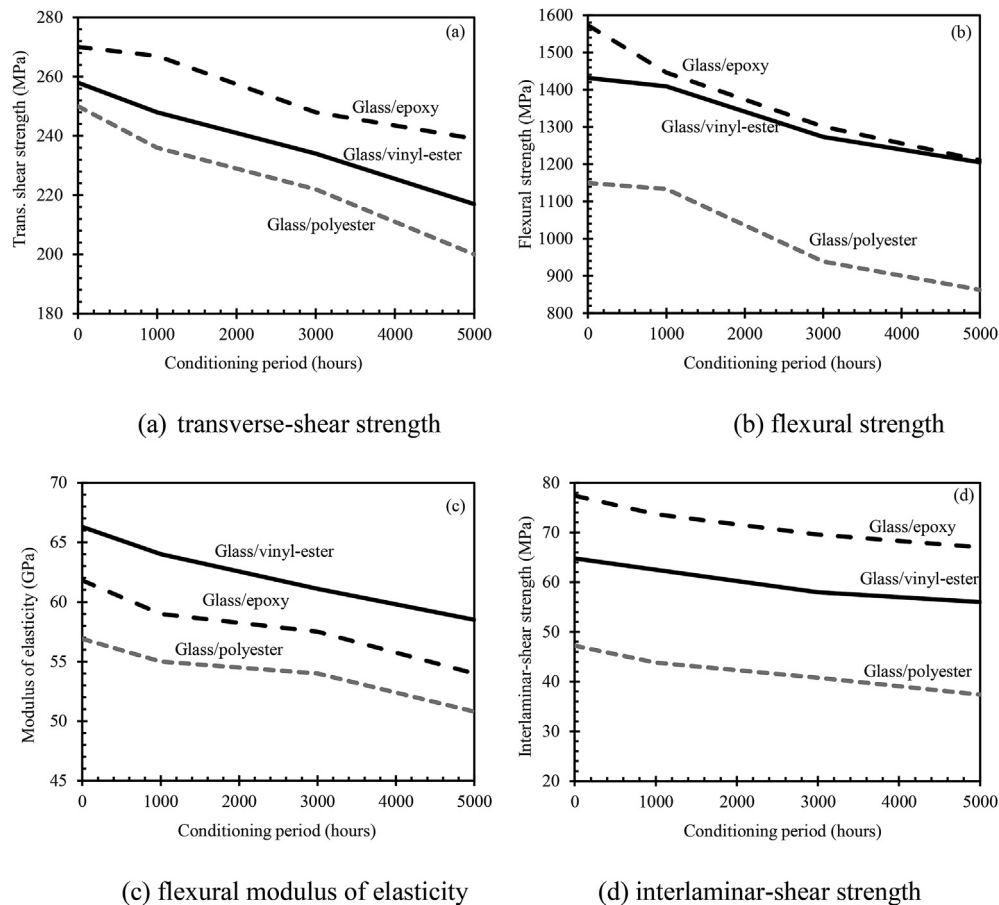
SEM observations were performed to investigate microstructural changes in the GFRP bars before and after conditioning. The specimens were cut, polished, and coated with a thin layer of gold/palladium in a vapor-deposit process. The analysis was carried out on a JEOL JSM-840 A microscope (JEOL, Akishima, Tokyo, Japan). Fig. 7 shows the SEM micrographs of the cross section of the reference GFRP bars, while Figs. 8–10 provide the SEM micrographs of the 5000 h conditioned specimens.

SEM analysis of the reference and conditioned specimens (Figs. 7–10) indicates that the GFRP bars made with vinyl-ester and epoxy evidenced no significant changes, but presented a slight

Table 4

Retention of mechanical properties of the conditioned GFRP bars.

Fiber/resin	Conditioned period	τ_u (MPa)	Retention (%)	f_u (MPa)	Retention (%)	E (GPa)	Retention (%)	S_u (MPa)	Retention (%)
Glass/polyester	1000	236	94.4	1133	99	55.0	96.6	43.8	93
	3000	222	88.8	939	81	54.0	94.9	40.8	87
	5000	194	77.5	863	75	50.8	89.3	37.4	79
Glass/vinyl-ester	1000	248	96.1	1409	98	64.0	96.5	62.5	97
	3000	234	90.7	1273	89	61.1	92.2	58.0	90
	5000	217	84.1	1186	83	58.5	88.2	56.0	87
Glass/epoxy	1000	267	98.9	1446	92	59.0	95.5	73.7	96
	3000	248	92.0	1301	83	57.5	93.0	69.6	90
	5000	239	89.0	1211	77	54.0	87.4	67.0	87

**Fig. 6.** Effect of conditioning in the alkaline solution at 60 °C on mechanical properties.

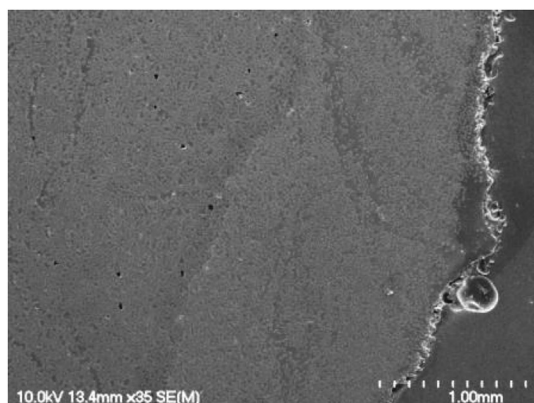
debonding at the interface between the fibers and vinyl-ester resin. Consequently, the vinyl-ester GFRP bars evidenced higher moisture uptake measured at saturation compared to the epoxy GFRP bars. On the other hand, the GFRP bars containing the polyester resin evidenced significant impact on the coating with the 5000 h conditioning. Moreover, these bars experienced greater debonding at the fiber–resin interface than did the vinyl-ester and epoxy GFRP bars. Accordingly, the polyester GFRP bars had higher moisture uptake measured at saturation and higher degradation rate of mechanical properties after conditioning.

SEM was also performed on the fracture zones of the 1000 h specimens after short-beam testing (Fig. 11) to investigate the mechanisms of failure at the interface fiber–matrix. The fiber surface of the vinyl-ester and epoxy GFRP bars had more resin coverage (Fig. 11 [b and c]) than the polyester GFRP bars (Fig. 11(a)).

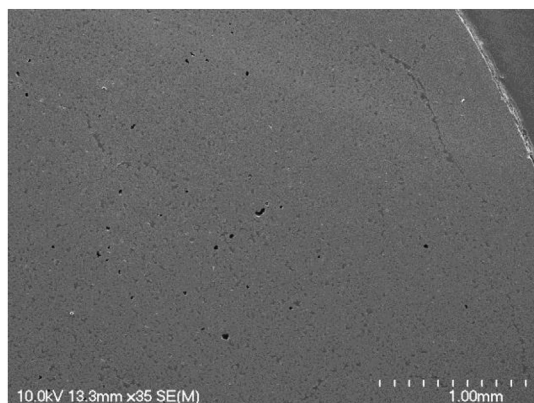
This observation corroborates the reduction ratio of the interlaminar-shear strength and flexural strength after conditioning in the alkaline solution and characterizes the higher bonding of the glass fiber to the vinyl-ester and epoxy resins than the polyester resin.

4.3.7. Glass transition temperature and cure ratio of the conditioned GFRP bars

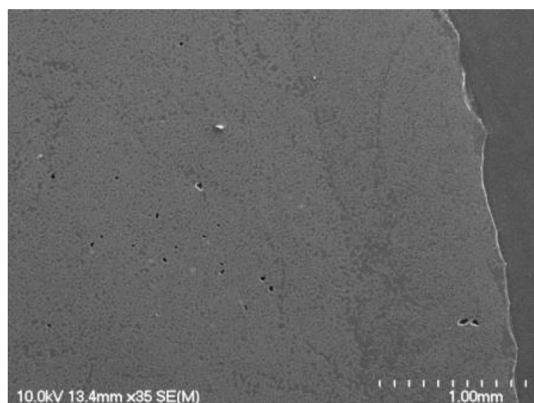
Differential scanning calorimetry (DSC) is used to obtain information about the thermal behavior and characteristics of polymeric materials and composites, such as the glass transition temperature (T_g) and curing process. In this study, 30–50 mg specimens from both unconditioned and conditioned specimens were sealed in aluminum pans and heated in a TA Instruments (New Castle, Delaware) DSC Q10 calorimeter to 200 °C at a rate of 20 °C/min. The



(a) Polyester GFRP bar



(b) Vinyl-ester GFRP bar



(c) Epoxy GFRP bar

Fig. 7. Micrographs of the cross section of the reference GFRP bars.

glass transition temperature (T_g) was determined in accordance with ASTM D3418 [21].

Table 5 presents the T_g values for the reference and specimens conditioned for 5000 h. The T_g of the conditioned polyester GFRP bars were slightly higher than that of the reference specimens, as a result of post-curing at high temperature. The vinyl-ester and epoxy GFRP bars were almost fully cured (99.1% and 100%, respectively); their T_g values were lower than that of the reference specimens by 11.5% and 10.3%, respectively. Epoxy resin is known to lower T_g when water is absorbed (plasticizing effect). The water absorption of the epoxy GFRP bars was 0.2%.

4.3.8. Chemical changes in the conditioned GFRP bars

Fourier transform infrared spectroscopy (FTIR) was used to

identify any chemical change/degradation after 5000 h of conditioning at 60 °C. FTIR spectra of the surface and core of the material specimens were recorded using a Jasco 4600 spectrometer equipped with an attenuated total-reflectance device. Five hundred and twelve scans were routinely acquired at a resolution of 4 cm^{-1} . Chemical degradation in the alkaline solution is mainly due to a hydrolysis reaction, which forms new hydroxyl ($-\text{OH}$) groups from sensitive units, such as ester groups. Hydroxyl groups appeared as a broad peak between 3200 and 3650 cm^{-1} , which corresponds to the stretching mode of the hydroxyl groups in the polyester, vinyl-ester, and epoxy resins.

Fig. 12 shows the FTIR spectra of the unconditioned and conditioned polyester, vinyl-ester, and epoxy GFRP specimens conditioned in the alkaline solution for 5000 h at 60 °C. For each specimen—reference and conditioned—spectra of the surface and core of the specimen were recorded and the areas of the O–H and C–H peaks calculated as presented in Fig. 13. Table 6 presents the ratio of the (OH^-) peak to the resin's carbon–hydrogen (C–H) stretching peak. The table indicates that none of the hydroxyl peaks for any of the tested vinyl-ester and epoxy GFRP specimens evidenced any significant changes, which equates to no increase in the amount of hydroxyl groups in the resins. This observation shows that the vinyl-ester and epoxy resins used did not degrade chemically while the specimens were immersed at 60 °C for 5000 h. On the other hand, the polyester resin showed significant differences on the surface and in the core of the tested specimens (see Table 6). The experimental O–H/C–H for the core and surface of the vinyl-ester and epoxy GFRP bars immersed for 5000 h were 1.5, 1.8, 1.2, and 1.5, respectively, compared to 1.80, 2.40, 1.25, and 1.6 for the unconditioned specimens, while the experimental ratios for the core and surface of the polyester GFRP bars immersed for 5000 h at 60 °C were 3.5 and 14.30, respectively. These results led to the conclusion that chemical degradation of the polymer occurred on the surface of the polyester bars, which was in direct contact with the solution during immersion. This observation explains the losses in mechanical properties of the polyester GFRP bars.

4.3.9. Moisture uptake at saturation of the conditioned GFRP bars

The moisture uptake at saturation of the reference and conditioned FRP bars was determined according to ASTM D570 [18]. The gain in mass was corrected to account for specimen mass loss due to possible dissolution phenomena. This correction was achieved by completely drying the immersed specimens in an oven at 100 °C for 24 h and comparing their masses to their initial masses. The mass loss may have occurred due to several causes: dissolution of soluble chemicals present on the surface; sand debonding in the case of sand-coated bars; and chemical degradation of one of the components, such as resin hydrolysis. In this study, the moisture-uptake ratios at saturation for the reference specimens were 1.15%, 0.63%, and 0.23% for the polyester, vinyl-ester, and epoxy GFRP, respectively, while these ratios for the conditioned specimens were 1.36%, 0.38%, and 0.20% for the polyester, vinyl-ester, and epoxy GFRP bars, respectively. The epoxy GFRP bars had the lowest water uptake, which is consistent with the lowest degradation of the fiber–resin interface as determined by DSC analysis and SEM observations.

5. Summary and conclusions

This study investigated glass-fiber-reinforced polymers (GFRPs) with polyester, vinyl-ester, and epoxy resins. Based on the results, the following conclusions concerning the glass FRP bars made with polyester, vinyl-ester and epoxy resins tested herein can be drawn:

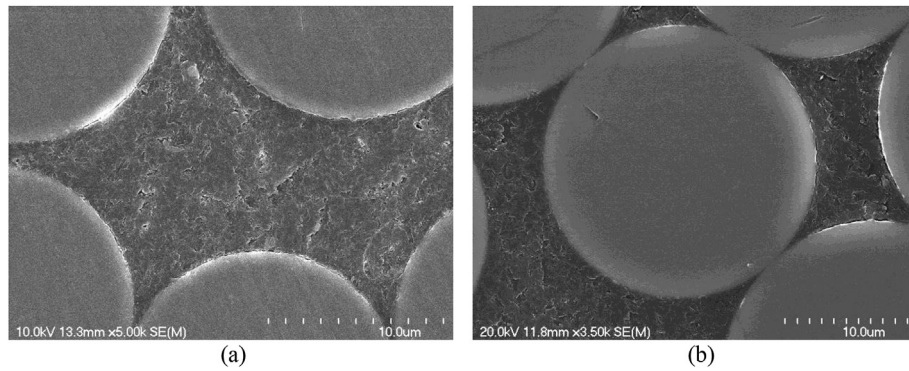


Fig. 8. Micrographs of the fiber–matrix interface of an epoxy GFRP bar before and after conditioning: (a) before conditioning; (b) after conditioning.

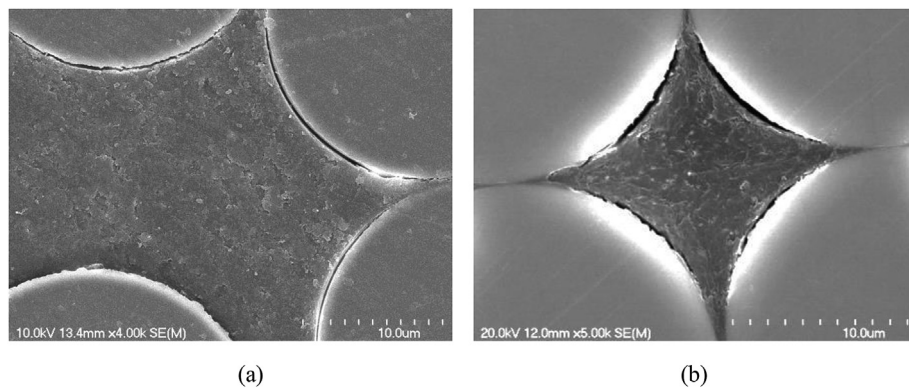


Fig. 9. Micrographs of the fiber–matrix interface of a polyester GFRP bar before and after conditioning: (a) before conditioning; (b) after conditioning.

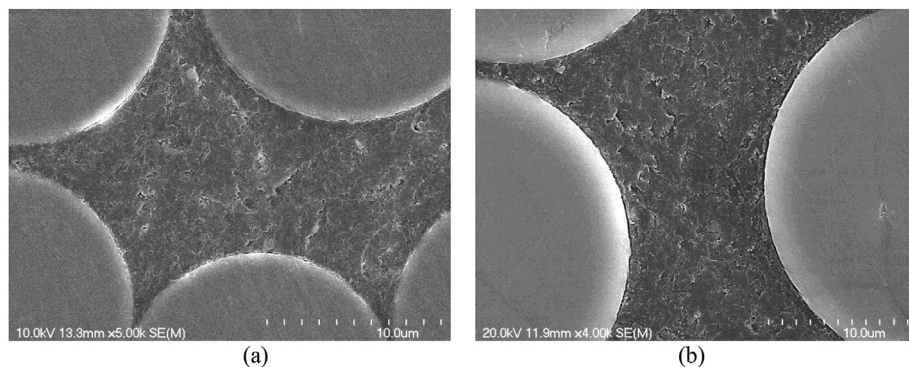


Fig. 10. Micrographs of the fiber/matrix interface of a vinyl-ester GFRP bar before and after conditioning: (a) before conditioning; (b) after conditioning.

1. The epoxy and vinyl-ester GFRP bars exhibited higher fiber–resin bond; flexural strength; flexural modulus of elasticity; and interlaminar-shear strength, which is governed by the fiber–matrix interface. In addition, they showed lower moisture uptake.
2. Both the polyester and epoxy GFRP bars had similar flexural-strength reductions after 5000 h of immersion (25% and 23%, respectively), while the vinyl-ester GFRP bars returned a lower reduction of 17%. These observations confirm that the bond between the GFRP fibers and polyester resin—before and after conditioning—was lower than that between the glass fibers and the vinyl-ester or epoxy resin.
3. The unconditioned polyester GFRP bars exhibited lower transverse-shear strength, flexural strength, interlaminar-shear strength, and the weakest fiber–resin interface. The transverse-shear strength of the polyester GFRP bars was significantly affected by accelerated aging (22.5% reduction after 5000 h), while the epoxy and vinyl-ester GFRP bars were slightly affected by accelerated aging (11% and 15.9% reductions, respectively, after 5000 h).
4. The flexural strength of the polyester GFRP bars was significantly affected by accelerated aging (25% reduction after 5000 h), while the vinyl-ester and epoxy GFRP bars were affected by accelerated aging (17% and 23% reductions, respectively, after 5000 h).
5. The interlaminar-shear strength of the polyester GFRP bars was highly affected by accelerated aging (21% reduction after 5000 h), while the vinyl-ester and epoxy GFRP bars were slightly

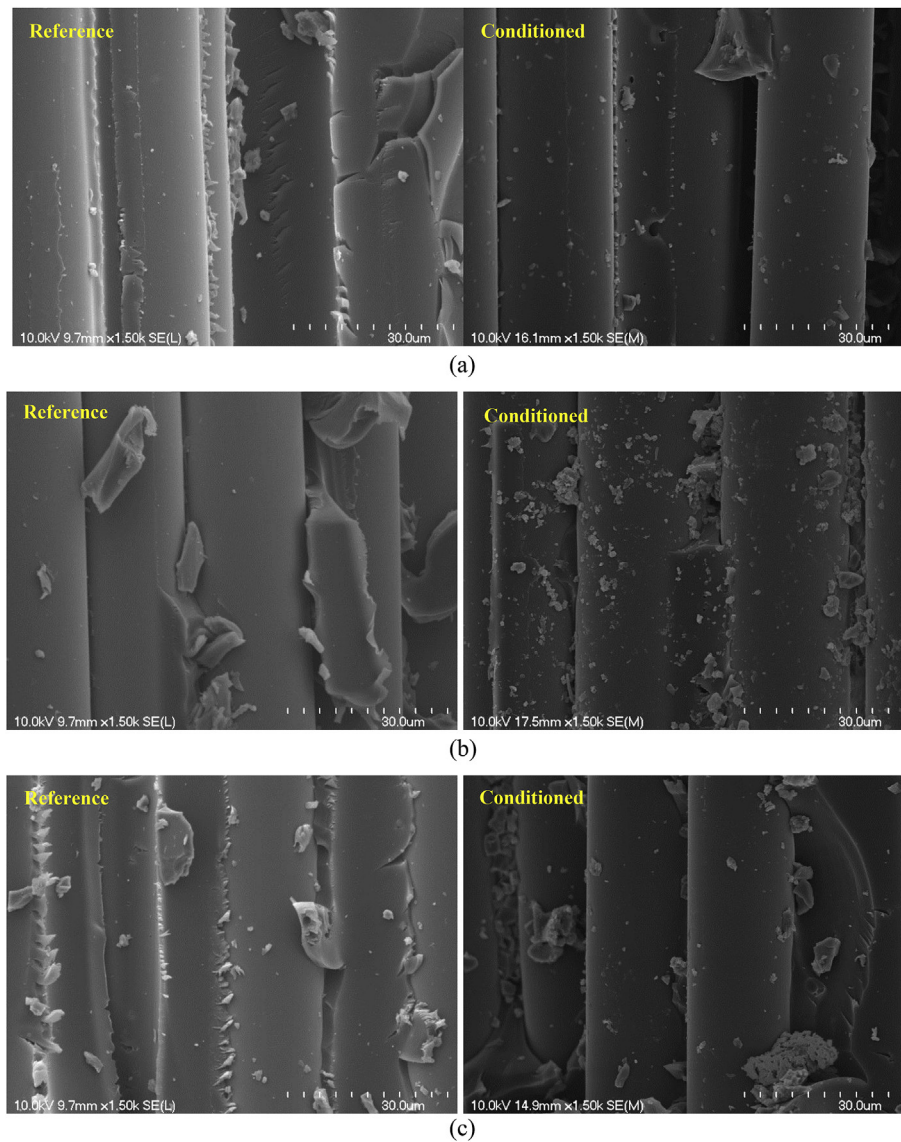


Fig. 11. Micrographs of bars conditioned in the alkaline solution for 1000 h at 60 °C (after interlaminar shear failure): (a) polyester GFRP; (b) vinyl-ester GFRP; (c) epoxy GFRP.

Table 5

Cure ratio, T_g , and moisture uptake of the reference and conditioned GFRP bars.

Property	GFRP bar type					
	Polyester		Vinyl-ester		Epoxy	
	Reference	5000 h	Reference	5000 h	Reference	5000 h
Cure ratio (%)	98	100	99	99	100	100
T_g (°C)	93	98	113	100	126	112
Moisture uptake (%)	1.15	1.36	0.63	0.38	0.23	0.20

- affected by accelerated aging (13% reduction each after 5000 h). The fiber–resin interface plays a significant role in controlling the degradation due to conditioning.
6. The microstructural observations revealed that GFRP bars made with vinyl-ester or epoxy resin were not significantly changed, but presented a slight debonding at the interface between the fibers and vinyl-ester resin. Consequently, the vinyl-ester GFRP bars evidenced higher moisture uptake measured at saturation compared to the epoxy GFRP bars.

7. The debonding at the interface between the fibers and polyester resin was higher than in the vinyl-ester and epoxy GFRP bars. Accordingly, the polyester GFRP bars evidenced higher moisture uptake measured at saturation and a higher degradation rate of mechanical properties after conditioning.
8. The polyester GFRP bars showed an increase in T_g of about 5 °C after conditioning due to post-curing (cure ratio of the reference specimens was 98.1%). The vinyl-ester and epoxy GFRP bars, however, experienced a decrease in T_g after conditioning.

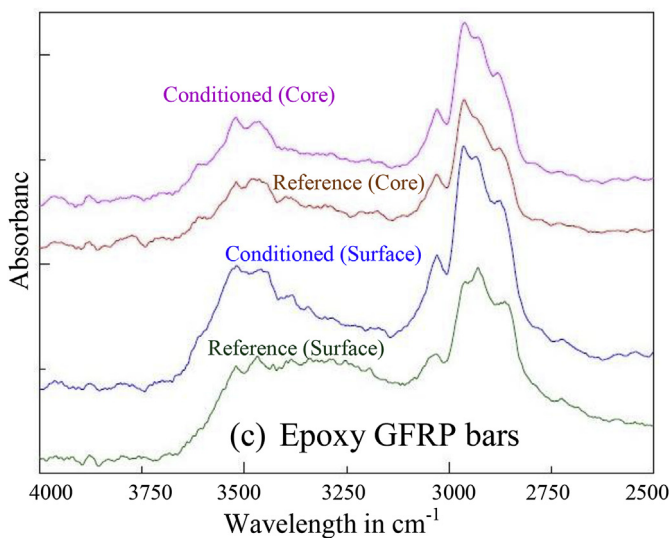
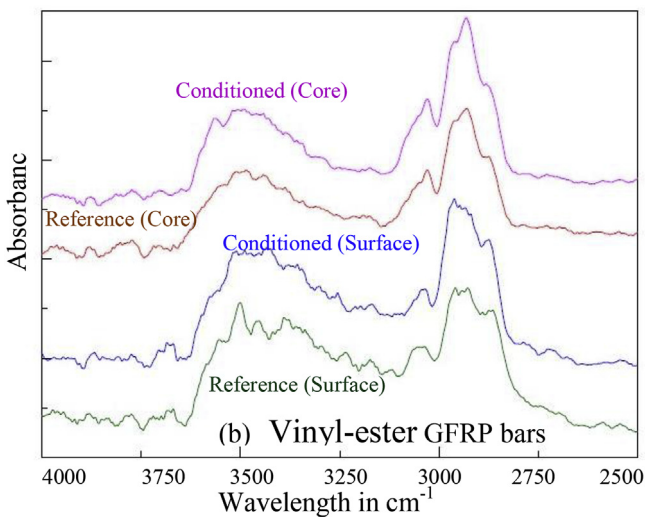
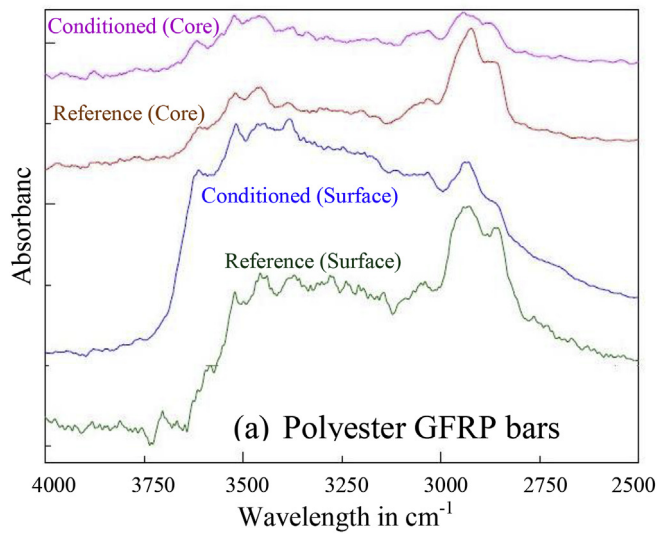


Fig. 12. FTIR spectra of reference and specimens conditioned for 5000 h.

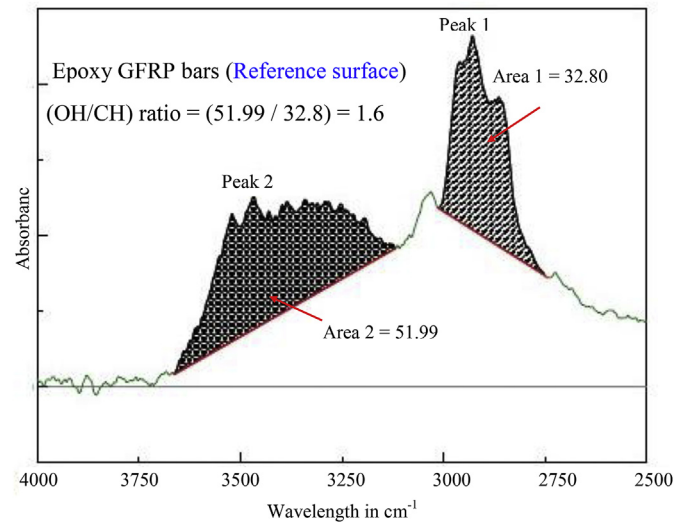


Fig. 13. Peak areas used to calculate a O–H/C–H.

Table 6
Ratio of the FTIR peaks.

Test location	OH/CH ratio					
	Polyester		Vinyl-ester		Epoxy	
	Reference	5000 h	Reference	5000 h	Reference	5000 h
Surface	2.60	14.3	2.40	1.80	1.60	1.50
Core	1.60	3.50	1.80	1.50	1.25	1.20

9. The polyester GFRP bars absorbed 18% more water than the vinyl-ester and epoxy GFRP bars after conditioning compared to the reference specimen.

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